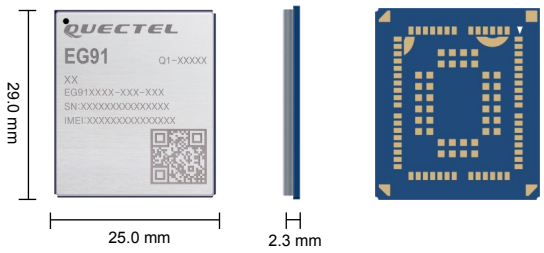
Quectel EG91 Series

IoT/M2M-optimized LTE Cat 1 Modules



Quectel EG91 is a series of LTE category 1 modules optimized specially for M2M and IoT applications. Adopting 3GPP Rel.11 LTE technology, it delivers maximum data rates up to 10 Mbps downlink and 5 Mbps uplink. These make EG91 series an ideal solution for numerous IoT applications that are not reliant on high speed connectivity but require the longevity and reliability of LTE network.

EG91 series contains 7 variants: EG91-NA, EG91-NS , EG91-VX, EG91-NAX, EG91-E, EG91-EX and EG91-AUX. EG91 series is pin-to-pin compatible with Quectel GSM M95 module, UMTS/HSPA UG95/UG96 modules and LPWA BG95 series/BG96/BC95-G modules. It is backward-compatible with existing GSM/GPRS and UMTS/HSPA networks, ensuring that it can be connected even in remote areas devoid of 4G network coverage.

A rich set of Internet protocols, industry-standard interfaces and abundant functionalities (USB serial drivers for Windows 7/8/10, Linux and Android, etc.) extend the applicability of the module to a wide range of M2M and IoT applications such as smart metering, wearable devices, environmental monitoring, asset tracking, fleet management, security and alarm systems and so on.

Key Benefits

- Cost-effective and low-power LTE connectivity optimized for broadband IoT applications
- Multi-band LTE, UMTS/HSPA and GSM/GPRS/EDGE coverage
- Compact SMT form factor ideal for size-constrained applications with extended operation temperature range
- Embedded power management unit (PMU) featuring ultra-low deep-sleep current consumption
- Simple migration from 2G/3G to 4G with a flexible and scalable platform

LTE
LTE Cat 1
Max 10 Mbps (DL)
Max 5 Mbps (UL)

HSPA+
Max 42 Mbps (DL)
Max 5.76 Mbps (UL)

LGA
LGA Package

Embedded Abundant Protocols

Compact Size

AT
Quectel Enhanced AT Commands

USB 2.0 High Speed Interface

USB Drivers

Rev.: V2.0 | Status: Released

Quectel EG91 Series

LTE Cat 1	EG91-NA	EG91-NS	EG91-VX	EG91-NAX	EG91-E	EG91-EX	EG91-AUX
Region/Operator	North America	North America	North America	North America	EMEA	EMEA	Latin America/ Australia/New Zealand
Dimensions (mm)	29.0 × 25.0 × 2.3	29.0 × 25.0 × 2.3	29.0 × 25.0 × 2.3	29.0 × 25.0 × 2.3	29.0 × 25.0 × 2.3	29.0 × 25.0 × 2.3	29.0 × 25.0 × 2.3
Temperature Range							
Operating Temperature	-35 to +75 °C	-35 to +75 °C	-35 to +75 °C	-35 to +75 °C	-35 to +75 °C	-35 to +75 °C	-35 to +75 °C
Extended Temperature	-40 to +85 °C	-40 to +85 °C	-40 to +85 °C	-40 to +85 °C	-40 to +85 °C	-40 to +85 °C	-40 to +85 °C
Frequency Bands							
LTE-FDD	B2/B4/B5/B12/B13	B2/B4/B5/B12/B13/ B25/B26	B4/B13	B2/B4/B5/B12/B13/ B25/B26	B1/B3/B7/B8/B20/ B28A	B1/B3/B7/B8/B20/B28	B1/B2/B3/B4/B5/B7/ B8/B28/B66
WCDMA	B2/B4/B5	B2/B4/B5	/	B2/B4/B5	B1/B8	B1/B8	B1/B2/B5/B8
GSM/EDGE	/	/	/	/	B3/B8	B3/B8	B2/B3/B5/B8
GNSS	GPS/GLONASS/BeiDou /Galileo/QZSS (Optional)	GPS/GLONASS/BeiDou /Galileo/QZSS (Optional)	GPS/GLONASS/BeiDou /Galileo/QZSS (Optional)	GPS/GLONASS/BeiDou /Galileo/QZSS (Optional)	/	GPS/GLONASS/BeiDou /Galileo/QZSS (Optional)	GPS/GLONASS/BeiDou /Galileo/QZSS (Optional)
Certifications							
Carrier	**America:** Verizon/AT&T/ T-Mobile/ U.S. Cellular **Canada:** Rogers/Telus	**America:** Verizon/AT&T/Sprint/ T-Mobile*	**America:** Verizon*	**America:** Sprint/Verizon*/ AT&T*/ T-Mobile*/ U.S. Cellular*	**Europe:** Deutsche Telekom	/	/
Regulatory	**Global:** GCF **America:** FCC/PTCRB **Canada:** IC	**Global:** GCF **America:** FCC/PTCRB **Canada:** IC	**Global:** GCF **America:** FCC	**Global:** GCF **America:** FCC/PTCRB **Canada:** IC	**Global:** GCF **Europe:** CE **Brazil:** Anatel **Australia & New Zealand:** RCM **Russia:** FAC	**Global:** GCF **Europe:** CE **Australia & New Zealand:** RCM	**Brazil:** Anatel* **Australia & New Zealand:** RCM*
Others	WHQL/RoHS	WHQL/RoHS	WHQL/RoHS	WHQL/RoHS	WHQL/RoHS	WHQL/RoHS	WHQL/RoHS
Data Transmission							
LTE-FDD Data Rate (Mbps)	10 (DL)/5 (UL)	10 (DL)/5 (UL)	10 (DL)/5 (UL)	10 (DL)/5 (UL)	10 (DL)/5 (UL)	10 (DL)/5 (UL)	10 (DL)/5 (UL)
DC-HSPA+ Data Rate (Mbps)	42 (DL)/5.76 (UL)	42 (DL)/5.76 (UL)		42 (DL)/5.76 (UL)	42 (DL)/5.76 (UL)	42 (DL)/5.76 (UL)	42 (DL)/5.76 (UL)
WCDMA Data Rate (kbps)	384 (DL)/384 (UL)	384 (DL)/384 (UL)		384 (DL)/384 (UL)	384 (DL)/384 (UL)	384 (DL)/384 (UL)	384 (DL)/384 (UL)
EDGE Data Rate (kbps)	/	/	/	/	296 (DL)/236.8 (UL)	296 (DL)/236.8 (UL)	296 (DL)/236.8 (UL)
GPRS Data Rate (kbps)	/	/	/	/	107 (DL)/85.6 (UL)	107 (DL)/85.6 (UL)	107 (DL)/85.6 (UL)
Interfaces							
(U)SIM	x 2	x 2	x 2	x 2	x 2	x 2	x 2
UART	x 2	x 2	x 2	x 2	x 2	x 2	x 2
USB 2.0	x 1	x 1	x 1	x 1	x 1	x 1	x 1
Audio Digital (PCM)	x 1	x 1	x 1	x 1	x 1	x 1	x 1
I2C	x 1	x 1	x 1	x 1	x 1	x 1	x 1
NETLIGHT	x 1	x 1	x 1	x 1	x 1	x 1	x 1
SPI	x 1	x 1	x 1	x 1	x 1	x 1	x 1
Main Antenna	x 1	x 1	x 1	x 1	x 1	x 1	x 1
Rx-diversity Antenna	x 1	x 1	x 1	x 1	x 1	x 1	/
GNSS Antenna	x 1	x 1	X 1	x 1	/	x 1	x 1
PWRKEY	x 1	x 1	x 1	x 1	x 1	x 1	x 1
RESET_N	x 1	x 1	x 1	x 1	x 1	x 1	x 1
USB_BOOT	x 1	x 1	x 1	x 1	x 1	x 1	x 1
Voice							
Speech Codec Modes	HR/FR/EFR/AMR/ AMR-WB	HR/FR/EFR/AMR/ AMR-WB	HR/FR/EFR/AMR/ AMR-WB	HR/FR/EFR/AMR/ AMR-WB	HR/FR/EFR/AMR/ AMR-WB	HR/FR/EFR/AMR/ AMR-WB	HR/FR/EFR/AMR/ AMR-WB
Echo Arithmetic	Echo Cancellation/ Noise Suppression	Echo Cancellation/ Noise Suppression	Echo Cancellation/ Noise Suppression	Echo Cancellation/ Noise Suppression	Echo Cancellation/ Noise Suppression	Echo Cancellation/ Noise Suppression	Echo Cancellation/ Noise Suppression
Audio	Digital Audio and VoLTE (Voice over LTE) (Optional)	Digital Audio and VoLTE (Voice over LTE) (Optional)	Digital Audio and VoLTE (Voice over LTE) (Optional)	Digital Audio and VoLTE (Voice over LTE) (Optional)	Digital Audio and VoLTE (Voice over LTE) (Optional)	Digital Audio and VoLTE (Voice over LTE) (Optional)	Digital Audio and VoLTE (Voice over LTE) (Optional)
Enhanced Features							
DTMF	●	●	●	●	●	●	●
DFOTA	●	●	●	●	●	●	●
QMI/RmNet	●	●	●	●	●	●	●
MIMO 2 x 2	●	●	●	●	●	●	/
Audio Playback*/ Audio Recording*	Optional	Optional	Optional	Optional	Optional	Optional	Optional
QuecOpen®	●	●	/	/	●	/	/
QuecLocator®	●	●	●	●	/	●	●
QuecFile	●	●	●	●	●	●	●
(U)SIM Card Detection	●	●	●	●	●	●	●

Copyright © 2020 Quectel Wireless Solutions Co., Ltd. All Rights Reserved http://www.quectel.com
HQ address: Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai, China 200233
Tel: +86 21 51086236 Email: info@quectel.com

Quectel EG91 Series

LTE Cat 1	EG91-NA	EG91-NS	EG91-VX	EG91-NAX	EG91-E	EG91-EX	EG91-AUX
Drivers							
USB Serial Driver	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x	Windows 7/8/8.1/10, Linux 2.6–5.4, Android 4.x–9.x
GNSS Driver	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x	/	Android 4.x–9.x	Android 4.x–9.x
RIL Driver	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x	Android 4.x–9.x
NDIS Driver	Windows 7/8/8.1/10	Windows 7/8/8.1/10	Windows 7/8/8.1/10	Windows 7/8/8.1/10	Windows 7/8/8.1/10	Windows 7/8/8.1/10	Windows 7/8/8.1/10
MBIM Driver	Windows 8/8.1/10, Linux 3.18–5.4	Windows 8/8.1/10, Linux 3.18–5.4	/	/	Windows 8/8.1/10, Linux 3.18–5.4	/	/
GobiNet Driver	Linux 2.6–5.4	Linux 2.6–5.4	Linux 2.6–5.4	Linux 2.6–5.4	Linux 2.6–5.4	Linux 2.6–5.4	Linux 2.6–5.4
QMI_WWAN Driver	Linux 3.4–5.4	Linux 3.4–5.4	Linux 3.4–5.4	Linux 3.4–5.4	Linux 3.4–5.4	Linux 3.4–5.4	Linux 3.4–5.4
Electrical Features							
Supply Voltage Range	3.3–4.3V, 3.8V Typ.	3.3–4.3V, 3.8V Typ.	3.3–4.3V, 3.8V Typ.	3.3–4.3V, 3.8V Typ.	3.3–4.3V, 3.8V Typ.	3.3–4.3V, 3.8V Typ.	3.3–4.3V, 3.8V Typ.
Power Consumption	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle	13 µA @ Power off 2.6 mA @ Sleep, Typ. 29 mA @ Idle

Notes:

1. * means under development.
2. ● means supported.

Copyright © 2020 Quectel Wireless Solutions Co., Ltd. All Rights Reserved http://www.quectel.com
HQ address: Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai, China 200233
Tel: +86 21 51086236 Email: info@quectel.com